

Homework 2

*Instructor: Henry Lam***Due Date:** Sep 26, 2025

Problem 1 Let $X \sim \text{Bin}(n, p)$ be a binomial random variable with parameters n and p . Recall that the p.m.f of X is given by

$$p(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n.$$

(a) Verify that

$$p(k+1) = \frac{n-k}{k+1} \cdot \frac{p}{1-p} \cdot p(k), \quad k = 0, 1, \dots, n-1.$$

- (b) Write an algorithm (or “pseudo code”) for the inverse transform method for generating X that utilizes the relation in part (a), in a way that you do not have to compute all the p.m.f. values, and also you only have to compute the c.d.f. values on the fly if needed.
- (c) Implement your answer in part (b) on a computer to generate 100 copies of $\text{Bin}(10, 2/3)$ and plot their distribution.

Problem 2 In this problem we will generate a negative binomial random variable with parameters r, p in three different ways. This random variable, called $NB(r, p)$, is the number of independent Bernoulli trials needed to get r successes, where each Bernoulli trial has success probability p . It has a p.m.f. given by

$$p(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, \dots$$

(a) Verify the relation

$$p(k+1) = \frac{k(1-p)}{k+1-r} p(k).$$

- (b) Use the relation in part (b) to give an algorithm for generating $NB(r, p)$.
- (c) Write down the relationship between $NB(r, p)$ and $\text{Geom}(p)$. Use it to obtain an algorithm to generate $NB(r, p)$.
- (d) Using the interpretation that $NB(r, p)$ counts the number of independent $\text{Ber}(p)$ trials required to accumulate r successes, obtain yet another approach for generating $NB(r, p)$.
- (e) Implement your algorithms in parts (a), (b) and (c) on a computer for generating $NB(2, 1/3)$. For each algorithm, generate 100 copies of X and plot their distribution. Are the plots similar?

Problem 3 Write an algorithm to generate the random variable X where

$$\mathbb{P}(X = j) = \left(\frac{1}{2}\right)^{j+1} + \left(\frac{1}{2}\right) \frac{2^{j-1}}{3^j}, \quad j = 1, 2, \dots$$

Implement your algorithm on a computer to generate 100 copies of X , and plot their distribution.

Problem 4 A fair die is to be continually rolled until all possible outcomes $1, 2, \dots, 6$ have occurred at least once, and we are interested in the total number of rolls in this experiment. Implement an algorithm on a computer to generate 100 copies of the total number of rolls, and plot the distributions of these 100 copies.